

Student Name:

BAULKHAM HILLS HIGH SCHOOL

Higher School Certificate

2011

Half-Yearly Examination

Mathematics

Extension 1

General Instructions

- Exam time – 1.5 hours and 5 mins reading time
- Start each question on a new page
- All necessary working should be shown, marks may be deducted for careless or badly arranged work.
- Write your student name at the top of each page of your answer booklet
- Board approved calculators may be used
- Write using black or blue pen

Total Marks: 72

Attempt ALL questions

Question 1 (9 marks)**Marks**

- a) Solve the inequality $x^2 - 5x < -6$ 2
- b) Factorise $x^3 - 4x + 5x^2y - 20y$ 2
- c) If $\cos \theta = \frac{-3}{7}$ and $\sin \theta > 0$, find an exact value for $\cot \theta$ 2
- d) Solve for $-90^\circ \leq x \leq 90^\circ$ to the nearest degree.
 $4 \sin^2 x - 1 = 0$ 3

Question 2 (9 marks) - Start a new page

- a) Express $4.0\dot{1}\dot{5}$ as a rational number showing all working. 2
- b) Factorise $8x^3 - (x - 1)^3$ 2
- c) Solve $\frac{4}{x-3} \geq 1$ 3
- d) A committee of 3 men and 5 women is to be formed from a group of 7 men and 8 women. Write an expression for the number of ways this can be done. 2

Question 3 (9 marks) - Start a new page

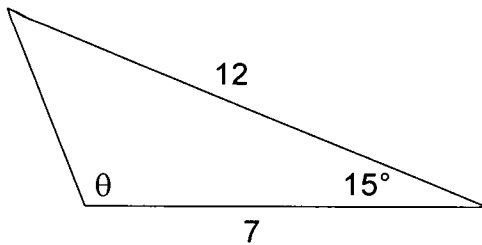
- a) How many nine-letter arrangements can be made using the letters of the word ISOSCELES? 2
- b) Find the number of ways in which 3 girls and 4 boys can be seated in a row so that the girls are next to each other. 2
- c) Find the greatest possible domain of
- (i) $x = 9 - y^2$ 1
- (ii) $y = \frac{2\sqrt{x+3}}{x-7}$ 2
- d) Prove the identity $\frac{\sin^3 x}{\cos x} + \sin x \cos x \equiv \tan x$ 2

Question 4 (9 marks) - Start a new page**Marks**a) Solve for $0^\circ \leq x \leq 360^\circ$ to the nearest degree.

(i) $\sin(x - 10^\circ) = 6 \cos(x - 10^\circ)$

2

(ii) $\sin(540^\circ - x) \cos x + \sin(90^\circ - x) = 0$

3b) Find θ to the nearest degree.**4****Question 5 (9 marks) - Start a new page**

a) Solve the simultaneous equations

$x + y = 2$ and

$2x^2 + xy - y^2 = 8$

2

b) Determine if the following functions are odd, even or neither (Show all working)

(i) $f(x) = x^3 - \sin x$

2

(ii) $f(x) = (x^3 - 2x)^{\frac{2}{3}}$

2c) Solve for x

$\sqrt{4x - 4} + \sqrt[4]{x - 1} = 3$

3**Question 6 (9 marks) - Start a new page**

a) Eight people are to be seated at a round table.

(i) How many seating arrangements are possible?

1(ii) Two people, Shelley and Hong, refuse to sit next to each other.
How many seating arrangements are then possible?**2**b) (i) On the same axes sketch the graphs of $y = |x|$ and $y = 6 - x^2$, clearly labelling an points of intersection.**3**(ii) Hence solve $|x| \geq 6 - x^2$ **1**c) The circle $x^2 + y^2 = 64$ and the parabola $y = ax^2 - b$ (where a and b are both positive) meet on both the x and y axes. Find a and b **2**

Question 7 (9 marks) - Start a new page

Marks

- a) Draw a neat sketch of

$$y = \frac{2x^2 - 50}{x - 5}$$

3

- b) A four digit number is to be made from the digits 2, 3, 4, 5, 7 and 9.
If no digit may be used more than once in the same number:

(i) how many four digit numbers can be made ?

1

(ii) how many ways can an odd four digit number can be made?

2

(iii) how many four digit numbers less than 3400 be made?

3

Question 8 (9 marks) - Start a new page

- a) If $f(x) = 9^x$ and if $f(x + 2) = kf(x)$, find k

2

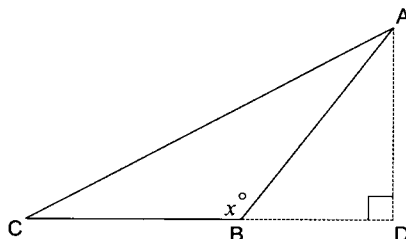
- b)

It is given that
 $3AD + BD = 2AB$

2

Show that:

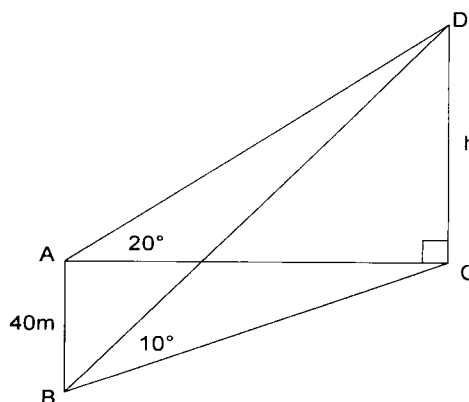
$$3 \sin x - \cos x = 2$$



- c)

The new Baulkham Hills High School pole, CD of height h metres, near the office, stands with base C on horizontal ground. A is a point on the ground due west of C and B is a point on the ground 40 metres due south of A.

From A and B the angles of elevation to the top of the pole are 20° and 10° respectively.



- (i) Show that $AC = \cot 20^\circ$ and find a similar expression for BC

2

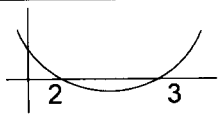
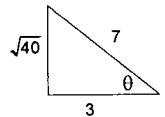
- (ii) Show that $h = \frac{40}{\sqrt{\cot^2 10^\circ - \cot^2 20^\circ}}$

2

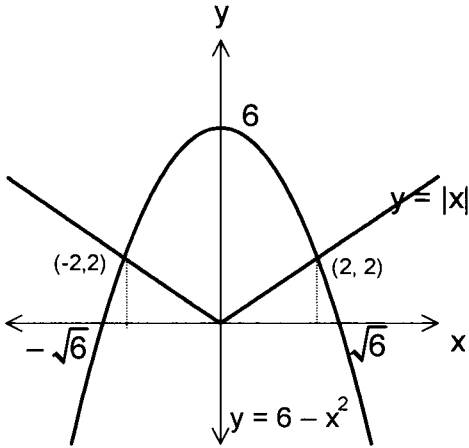
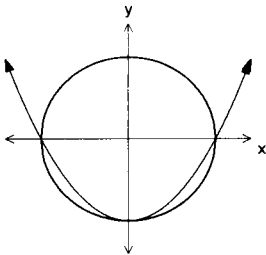
- (iii) Hence find the height of the pole to the nearest metre.

1

Yr 11 Ext 1 HALF YEARLY 2011 SOLUTIONS

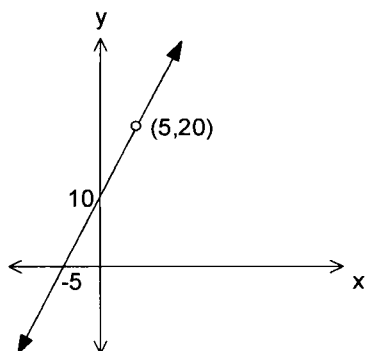
Question 1 (9 marks)		Mk	Comments
a)	$x^2 - 5x + 6 < 0$ $(x-3)(x-2) < 0$ $2 < x < 3$ 	2	<u>1 Mark</u> Correctly factorised
b)	$x^2 - 4x + 5x^2y - 20y$ $= [x(x^2 - 4) + 5y(x^2 - 4)]$ $= (x^2 - 4)(x + 5y)$ $= (x+2)(x-2)(x+5y)$	2	<u>1 Mark</u> Partially factorises
c)	 in the 2nd quadrant $\cot \theta = -\frac{7}{\sqrt{40}}$	2	<u>1 Mark</u> - Correctly identifies quadrant/sign - Finds $\sqrt{40}$
d)	$\sin^2 x = \frac{1}{4}$ $\sin x = \pm \frac{1}{2}$ Acute $x = 30^\circ$ $x = \pm 30^\circ$	3	<u>2 Mark</u> - Correct solution over another domain containing at least 2 answers <u>1 Mark</u> - Correct soln for $\sin x = \frac{1}{2}$ $\sin x = \pm \frac{1}{2}$
Question 2 (9 marks) - Start a new page			
a)	$10x = 40.\dot{1}\dot{5}$ $1000x = 4015.\dot{1}\dot{5}$ $990x = 3975$ $x = \frac{3975}{990} = \frac{265}{66} = 4\frac{1}{66}$	2	<u>1 Mark</u> - Uses a correct method - Not simplified
b)	$8x^3 - (x-1)^3$ $= [2x - (x-1)][4x^2 + 2x(x-1) + (x-1)^2]$ $= [x+1][7x^2 - 4x + 1]$	2	<u>1 Mark</u> - Correctly factorises $a^3 - b^3$ - Simplifies the factorisation - Finds one correct factor
c)	$\frac{4}{x-3} \geq 1 \quad x \neq 3$ $4(x-3) \geq (x-3)^2$ $0 \geq (x-3)^2 - 4(x-3)$ $(x-3)(x-3-4) \leq 0$ $(x-3)(x-7) \leq 0 \text{ but } x \neq 3$ $3 < x \leq 7$	3	<u>2 Marks</u> - Bald answer - Identifies 2 critical points via a correct method - Correct conclusion to their critical points obtained using a correct method <u>1 Mark</u> - Uses a correct method - Acknowledges $x \neq 3$ <u>0 Mark</u> - Solves like an eqn with no consideration of the denominator
d)	${}^7C_3 \times {}^8C_5 = 1960$	2	Unsimplified answer is ok <u>1 Mark</u> 7C_3 or 8C_5 involved in answer - Bald answer

Question 5 (9 marks) - Start a new page		
a)	$x + y = 2$ -----(1) $(x + y)(2x - y) = 8$ -----(2) Sub (1) $\rightarrow$ (2) $2x - y = 4$ -----(3) (3) + (1) $3x = 6$ <u>$x = 2$</u> Sub $x \rightarrow$ (1) $y = 0$ <u>$\therefore x = 2, y = 0$</u>	2 <u>1 Mark</u> - Finds x or y - Substitutes to find the second value correctly
b)	(i) $f(x) = x^3 - \sin x$ $f(-x) = (-x)^3 - \sin(-x)$ $= -x^3 + \sin x$ $= -(x^3 - \sin x)$ $= -f(x)$ $\therefore f(x)$ is odd (ii) $f(x) = (x^3 - 2x)^{\frac{2}{3}} = \sqrt[3]{(x^3 - 2x)^2}$ $f(-x) = \sqrt[3]{[(-x)^3 - 2(-x)]^2}$ $= \sqrt[3]{(-x^3 + 2x)^2}$ $= \sqrt[3]{(x^3 - 2x)^2}$ $= f(x)$ $\therefore f(x)$ is even	2 <u>(i) 1 Mark</u> Attempts to find $f(-x)$ 2 <u>(ii) 1 Mark</u> Attempts to find $f(-x)$
c)	$\sqrt{4x - 4} + \sqrt[4]{x - 1} = 3$ $2\sqrt{x - 1} + \sqrt[4]{x - 1} = 3$ $2(x - 1)^{\frac{1}{2}} + (x - 1)^{\frac{1}{4}} - 3 = 0$ Let $(x - 1)^{\frac{1}{4}} = a$ $2a^2 + a - 3 = 0$ $(2a + 3)(a - 1) = 0$ $a = -\frac{3}{2}$ or $a = 1$ $\sqrt{x - 1} = -\frac{3}{2}$ or $\sqrt{x - 1} = 1$ No soln or $x - 1 = 1$ <u>$\therefore x = 2$</u>	3 <u>2 Marks</u> - Reduces it to a quadratic <u>1 Mark</u> - Simplifies $\sqrt{4x - 4}$ - Bald answer <u>0 marks</u> - Simplified question to $(4x - 4)^2 + (x - 1) = 3$ [or equivalent]

Question 6 (9 marks) - Start a new page			
a)	(i) $7! = 5040$ (ii) sit together $= 6! \times 2 = 1440$ Do not sit together $= 7! - 6! \times 2 = 3600$	1	(i) Unsimplified answer is ok (ii) <u>1 Mark</u> - Correctly subtracts for total - Finds <i>sit together</i> $= 6! \times 2$
b)	(i)  Points of intersection <u>When $x \geq 0$</u> $x = 6 - x^2$ $x^2 + x - 6 = 0$ $(x - 2)(x + 3) = 0$ $x = 2$ or $x = -3$ (not in domain) <u>When $x < 0$</u> by symmetry $x = -2$ or $x = 3$ (not in domain) $\therefore$ points of intersection are $(2, 2)$ and $(-2, 2)$ (ii) <u>$x \leq -2$ or $x \geq 2$</u>	3	(i) <u>2 Marks</u> 2 correct graphs and important info 1 correct graph with pts of intersection <u>1 Mark</u> - One correct graph - Finds and shows the points of intersection.
c)	Parabola $y = a(x - 8)(x + 8)$ $y = ax^2 - 64a$ ----(1) $y - \text{int when } x = 0$ $-64a = -8$ $a = \frac{1}{8}$ Sub $a \rightarrow$ (1) $\therefore y = \frac{1}{8}x^2 - 8$ $\therefore a = \frac{1}{8}, b = 8$ 	2	<u>1 Mark</u> Finds a or b

Question 7 (9 marks) - Start a new page

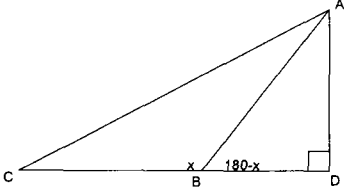
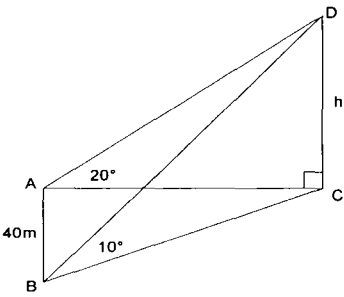
a) $y = \frac{2x^2 - 50}{x - 5}$
 $y = \frac{2(x + 5)(x - 5)}{x - 5}$
 $\therefore y = 2(x + 5)$ where $x \neq 5$
 ie. It is discontinuous at $(5, 20)$



- 3** 2 Marks
- Draws $y = x + 5$
 - Establishes function is equivalent to $y = x + 5$
- 1 Mark
- Identifies $x = 5$ is discontinuous

b) (i) $6 \times 5 \times 4 \times 3 = 360$
 (ii) $4 \times 5 \times 4 \times 3 = 240$
 (iii) Less than 3000 = $1 \times 5 \times 4 \times 3 = 60$
 Between 3000 and 3400 = $1 \times 1 \times 4 \times 3 = 12$
 Total = 72

- 1** (i) unsimplified is ok
- 2** (ii) 1 Mark
- Considers last digit is odd
- 3** (iii) 2 Mark
- Considers both cases (starting with 32__ and 2__)
- 1 Mark
- Considers one of the above cases

Question 8 (9 marks) - Start a new page		
a)	$f(x) = 9^x$ $f(x+2) = 9^{x+2}$ $f(x+2) = kf(x)$ $9^{x+2} = k \times 9^x$ $k = \frac{9^{x+2}}{9^x}$ $k = 9^2$ $k = 81$	2 <u>1 Mark</u> - Correctly finds $f(x+2)$ - Correctly finds k from their $f(x+2)$
b)	$\angle ABD = 180^\circ - x$ (straight line) $\sin(180^\circ - x) = \frac{AD}{AB}$ $\sin x = \frac{AD}{AB}$ $\cos(180^\circ - x) = \frac{BD}{AB}$ $-\cos x = \frac{BD}{AB}$ $\cos x = -\frac{BD}{AB}$ given that $3AD + BD = 2AB$ ----- (1) Show that: $3 \sin x - \cos x = 2$ $LHS = 3 \frac{AD}{AB} - \frac{BD}{AB}$ $= \frac{3AD + BD}{AB}$ $= \frac{2AB}{AB} \sim \text{from (1)}$ $= 2$ $= RHS$ 	2 <u>1 Mark</u> - Identifies $\sin(180^\circ - x) = \frac{AD}{AB}$ and $\cos(180^\circ - x) = \frac{BD}{AB}$ - Substitutes $\sin(180 - x) = \sin x$ or $\cos(180 - x) = -\cos x$
c)	i) $\tan 20^\circ = \frac{h}{AC}$ $AC = h \cot 10^\circ$ $\tan 10^\circ = \frac{h}{BC}$ $BC = h \cot 10^\circ$ (ii) Show that $h = \frac{40}{\sqrt{\cot^2 10^\circ - \cot^2 20^\circ}}$ $40^2 = BC^2 - AC^2$ (by Pythag) $40^2 = h^2 \cot^2 20^\circ - h^2 \cot^2 10^\circ$ $40^2 = h^2 (\cot^2 20^\circ - \cot^2 10^\circ)$ $h^2 = \frac{40^2}{\cot^2 20^\circ - \cot^2 10^\circ}$ $h = \frac{40}{\sqrt{\cot^2 20^\circ - \cot^2 10^\circ}}$ (iii) $h = 6.347 \dots$ $h = 6.35m$ 	2 (i) <u>1 Mark</u> - Shows AC - Writes a correct expression for BC 2 (ii) <u>1 Mark</u> - Uses Pythagoras in an attempt to show h or equivalent 1 (iii)
-- End of Exam --		